Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Engineering & Technology
B.Tech. | Sem-7 | Winter-2025

Course Code	:	2101CS701	**Date**	:	08-10-2025
Course Name	:	Compiler Design	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** Explain cousins of compiler. **4**

(B) Explain types of compiler. **3**

OR

What is pass and phases of compiler?

(C) Explain phases of compiler. **7**

OR

Differentiate interpreter and compiler.

Q.2 **(A)** Explain lexical analysis with suitable example. **4**

(B) Draw a DFA : (0+1)*110. **3**

OR

Draw a DFA for binary string contains 0110.

(C) Convert the following RE into DFA using Thomson's construction: (a + b)* bb **7**

OR

Convert the following RE into DFA: (a + b)* a* b

Q.3 **(A)** Check the given grammar is LL(1) or not? **4**

S→AaAb | BbBa
A→ε
B→ε

(B) Explain left recursion and left factoring with example. **3**

OR

Explain shift reduce parser.

(C) Construct SLR parsing table for the following grammar : **7**

E→E+T | T
T→T*F | F
F→id | (E)

OR

Construct syntax directed translation scheme that translate arithmetic operation from infix to postfix. Give annotated parse tree for the input: 5*3+2

Q.4	**(A)**	Write a different forms of Intermediate code generation for given input string: X= (a + b) * c / (d - e)	**4**
	(B)	Explain activation record.	**3**

OR

Explain call by value and call by reference with example.

	(C)	Translate following arithmetic expression - (a * b) + (c + d) - (a + b + c + d) into 1] Quadruples 2] Triple 3] Indirect Triple	**7**

OR

Compare Different Storage Allocation Strategies

Q.5	**(A)**	Explain Peephole Optimization in detail.	**4**
	(B)	Explain the partitioning method for the basic block.	**3**

OR

What is Symbol table? Discuss different data Structures suitable for it.

	(C)	Discuss generic issues in the design of code generation	**7**

OR

Explain various code optimization techniques.

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